BT-203 (GS)

B.Tech., I & II Semester Examination, June 2026

Grading System (GS)

BASIC MECHANICAL ENGINEERING

Time : Three Hours Maximum Marks : 70

Note : (i) Attempt any five questions. (ii) All questions carry equal marks. (iii) Original paper bilingual (Hindi + English) tha; yahan sirf English version diya gaya hai.

Coverage: Full paper (Q1–Q8)

1. (a) Explain the Iron–Carbon equilibrium diagram in detail. Discuss all phases, invariant reactions and critical temperatures. **7**

(b) Describe the composition, classification and properties of cast irons. **7**

2. (a) Derive Hooke's Law and explain the concept of modulus of elasticity. **7**

(b) Explain various methods of temperature measurement. **7**

3. (a) Explain the construction, working principle, least count and applications of Vernier callipers. **7**

(b) Explain the constructional features and working principle of a centre lathe. **7**

4. (a) Explain the classification of hydraulic turbines. Compare impulse and reaction turbines. **7**

(b) Explain the working principle, construction and applications of pumps. **7**

5. (a) Derive Bernoulli's Equation for incompressible fluid flow from first principles. Discuss the assumptions involved. **7**

(b) Explain the concept of a thermodynamic system. Classify thermodynamic systems into open, closed and isolated systems with suitable engineering examples. **7**

6. (a) State and explain the Zeroth Law of Thermodynamics. Discuss its significance in temperature measurement and thermal equilibrium. **7**

(b) Classify boilers based on construction, circulation, pressure and application. **7**

7. (a) Explain the construction and working principle of a steam engine. **7**

(b) Discuss the working principle of a two-stroke diesel engine and compare it with a two-stroke petrol engine. **7**

8. Write a short note on any two: **14**

(i) Working principle of compressors (ii) Alloy steels and their importance in modern engineering

(iii) Sine-bar (iv) Natural and artificial draught

****

Yeh reconstructed paper kya hai? RGPV ke 2022–2026 ke papers ka free official PDF internet par downloadable nahi hai (mirrors sirf ~2019 tak ke papers rakhte hain). Ye version un hi papers ke public web renders se word-by-word English questions lekar banaya gaya hai, tai ki students unhe PDF me padh/print kar sakein. Hindi version aur exact layout ke liye original paper dekhein. Ye unofficial study material hai.